

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

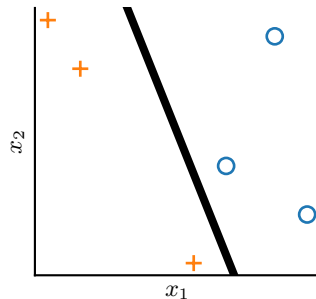
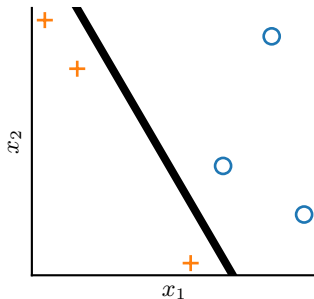
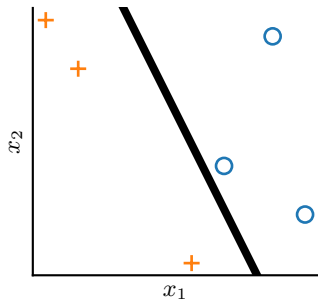
Slide Set 11: Support Vector Machines



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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 - ▶ For use only by enrolled students in the course

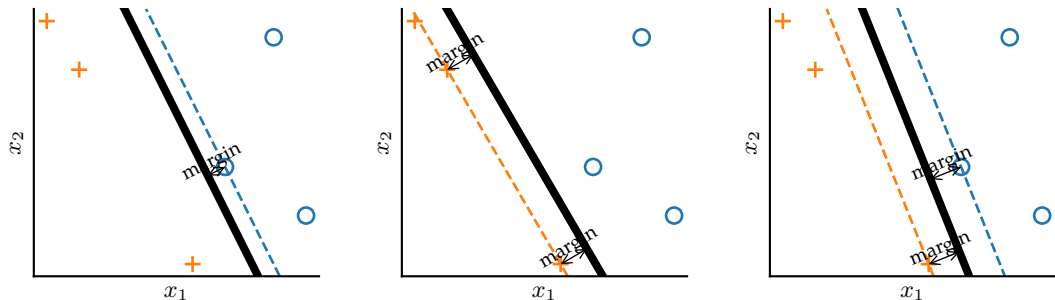
Margin

Consider these decision surfaces, each with zero training error.



Margin

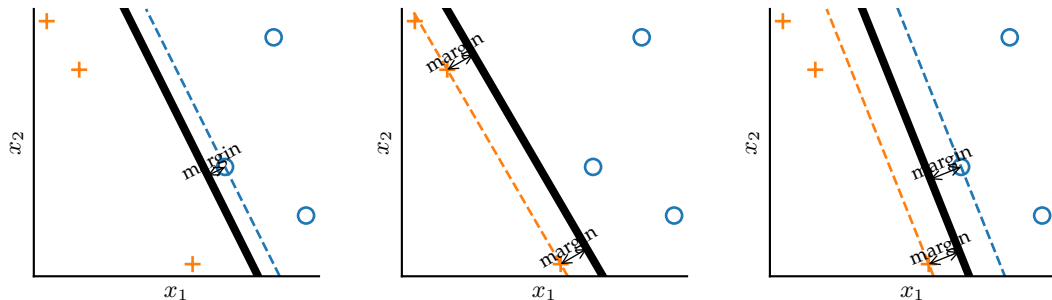
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The margin is the distance from the decision boundary to the closest point(s).

Margin

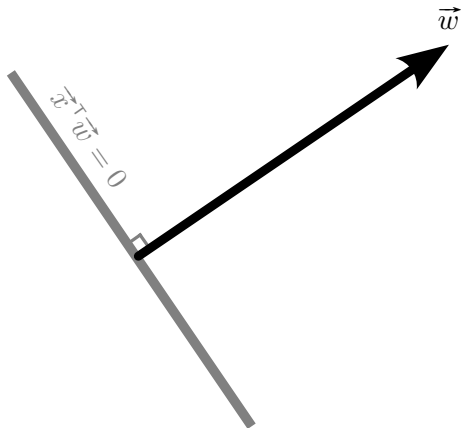
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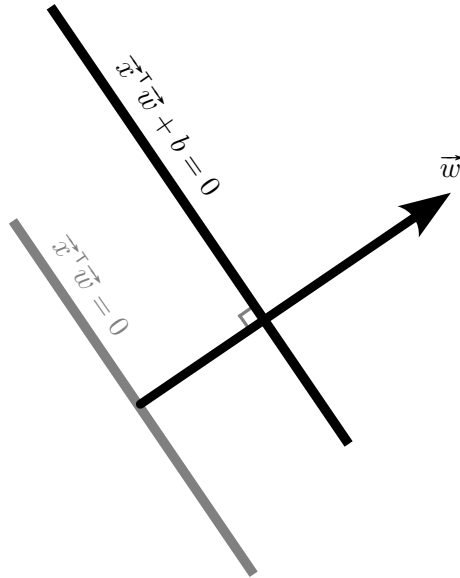
The margin is the distance from the decision boundary to the closest point(s).

How to find the (linear) separator that has the largest margin?

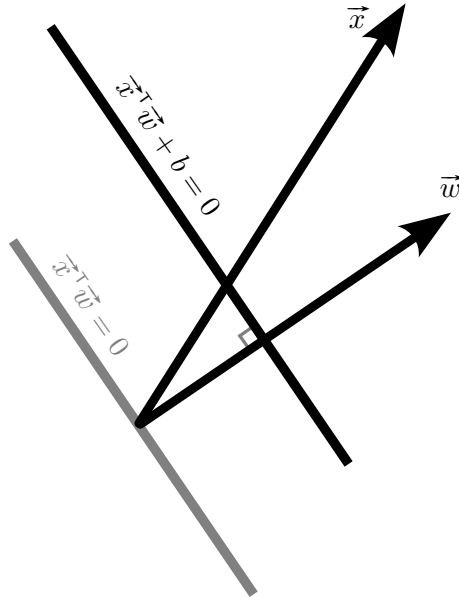
Margin of a Point



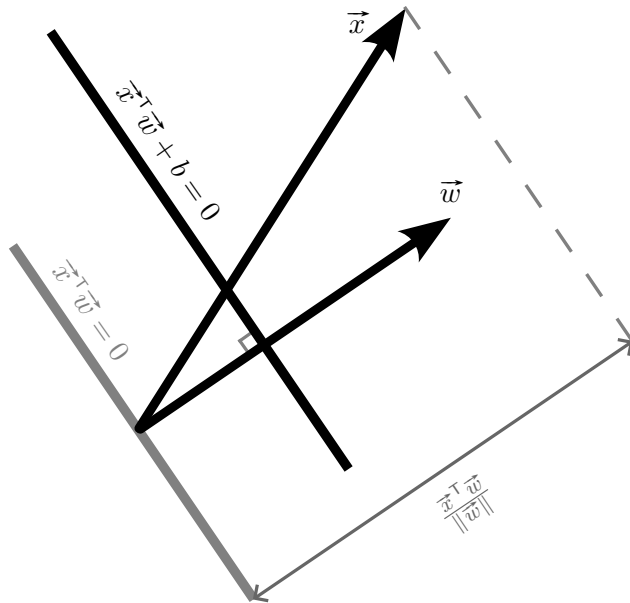
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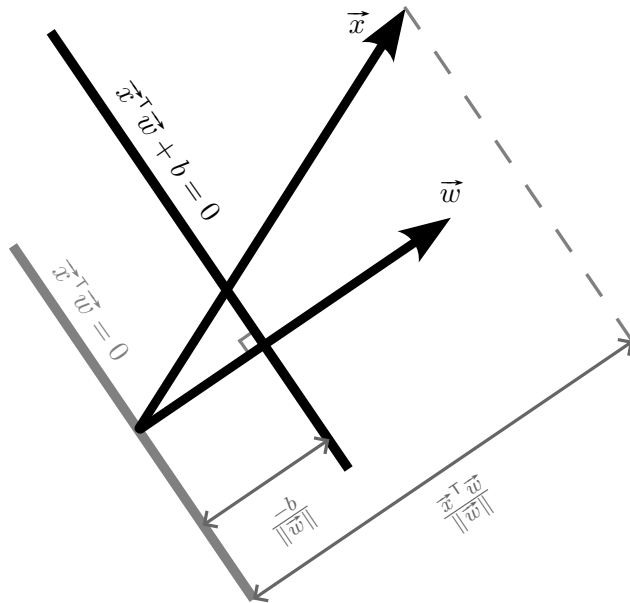
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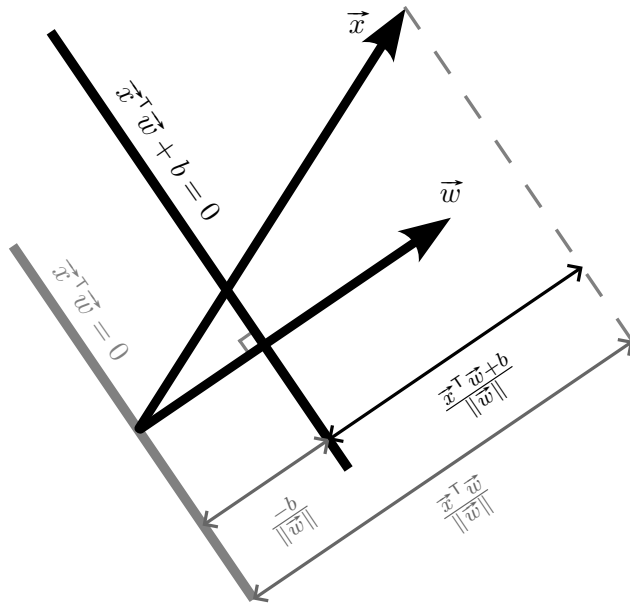
Margin of a Point



Margin of a Point



Margin of a Point



Margin of Data Point

Thus, $\frac{\vec{x}^T \vec{w} + b}{\|\vec{w}\|}$ is

- Distance from point to boundary,
- *But*, negative if the point is classified as negative.

Margin of Data Point

Thus, $\frac{\vec{x}^T \vec{w} + b}{\|\vec{w}\|}$ is

- Distance from point to boundary,
- *But*, negative if the point is classified as negative.

The **margin of a data point i** is

$$\frac{y_i (\vec{x}_i^T \vec{w} + b)}{\|\vec{w}\|} .$$

It is

- Distance from data point to boundary,
- *But*, negative if the point is incorrectly classified.

Margin Definitions

Data point i

$$\frac{y_i (\vec{x}_i^T \vec{w} + b)}{\|\vec{w}\|}$$

Margin Definitions

Data point i	Dataset
$\frac{y_i(\vec{x}_i^\top \vec{w} + b)}{\ \vec{w}\ }$	$\min_{i=1}^m \frac{y_i(\vec{x}_i^\top \vec{w} + b)}{\ \vec{w}\ }$

Margin Definitions

	Data point i	Dataset
Geometric	$\frac{y_i (\vec{x}_i^\top \vec{w} + b)}{\ \vec{w}\ }$	$\min_{i=1}^m \frac{y_i (\vec{x}_i^\top \vec{w} + b)}{\ \vec{w}\ }$
Functional	$y_i (\vec{x}_i^\top \vec{w} + b)$	$\min_{i=1}^m y_i (\vec{x}_i^\top \vec{w} + b)$

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Functional	$y_i (\vec{x}_i^\top \vec{w} + b)$	$\min_{i=1}^m y_i (\vec{x}_i^\top \vec{w} + b)$

Geometric margin: (signed) distance, multiplied by y

Functional margin: value of discriminant, multiplied by y

Data point: margin of that data point

Dataset: smallest value across entire dataset

Maximum Margin Formulation

How to maximize $\min_{i=1}^m \frac{y_i (\vec{x}_i^\top \vec{w} + b)}{\|\vec{w}\|}$?

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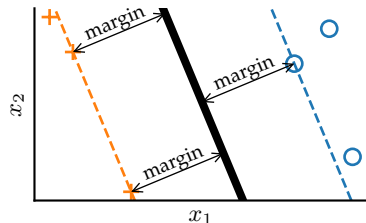
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To remove extra degree of freedom: set *functional margin* of closest point(s) to be equal to 1. Then, the *geometric margin* is $\frac{1}{\|\vec{w}\|}$.

Thus, maximizing the margin is the same as

- minimizing $|\vec{w}|$ (or minimizing $|\vec{w}|^2 = \vec{w}^\top \vec{w}$)
- if the functional margin of the closest point is 1



The maximum margin for this dataset

Maximum Margin Quadratic Program

$$\begin{aligned} \arg \min_{\vec{w}, b} \quad & \frac{1}{2} \vec{w}^\top \vec{w} \\ \text{s.t.} \quad & y_i (\vec{x}_i^\top \vec{w} + b) \geq 1 \quad \text{for all } i \in \{1, \dots, m\} \end{aligned}$$

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This is a quadratic program (QP). It has

- A *quadratic* objective: $\frac{1}{2} \vec{w}^\top \vec{w}$
- *Linear* constraints: $y_i (\vec{x}_i^\top \vec{w} + b) \geq 1$
(one for each example in the training set)

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- *Linear* constraints: $y_i (\vec{x}_i^\top \vec{w} + b) \geq 1$ — ... which makes this equality for at least one point (one for each example in the training set)
... causes this to be as small as possible ...

Solving Maximum Margin QP

General QP formulation:

$$\begin{aligned} \arg \min_{\vec{z}} \quad & \frac{1}{2} \vec{z}^\top \mathbf{P} \vec{z} + \vec{q}^\top \vec{z} \\ \text{s.t.} \quad & \mathbf{G} \vec{z} \leq \vec{h} \end{aligned} \quad \left(\text{each element of } \mathbf{G} \vec{z} \text{ is } \leq \text{than the same element of } \vec{h} \right)$$

Our QP:

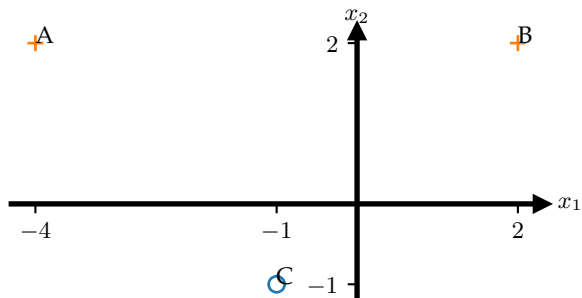
$$\begin{aligned} \arg \min_{\vec{w}, b} \quad & \frac{1}{2} \vec{w}^\top \vec{w} \\ \text{s.t.} \quad & y_i (\vec{x}_i^\top \vec{w} + b) \geq 1 \end{aligned} \quad \text{for all } i \in \{1, \dots, m\}$$

To make them match:

$$\vec{z} = \begin{bmatrix} b \\ \vec{w} \end{bmatrix} \quad \mathbf{P} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & & \\ 0 & 0 & 1 & & \\ \vdots & & & \ddots & \\ 0 & \dots & \dots & 0 & 1 \end{bmatrix} \quad \vec{q} = \vec{0} \quad \mathbf{G} = - \begin{bmatrix} y_1 & \text{---} y_1 \vec{x}_1^\top \text{---} \\ y_2 & \text{---} y_2 \vec{x}_2^\top \text{---} \\ \vdots & \vdots \\ y_m & \text{---} y_m \vec{x}_m^\top \text{---} \end{bmatrix} \quad \vec{h} = -\vec{1}$$

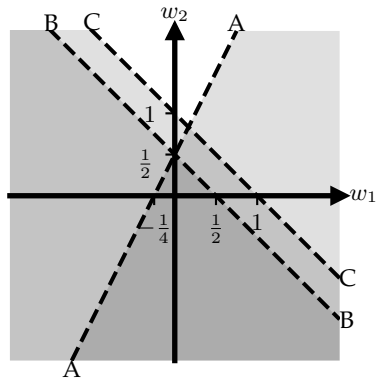
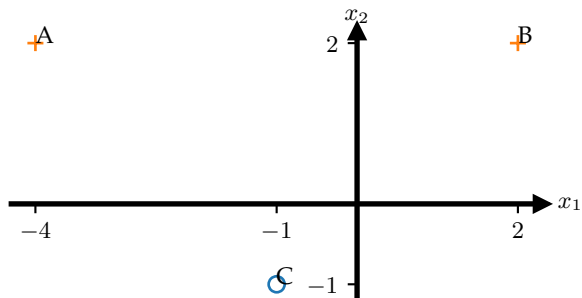
Maximum Margin Example

Simple example with b forced to be 0:



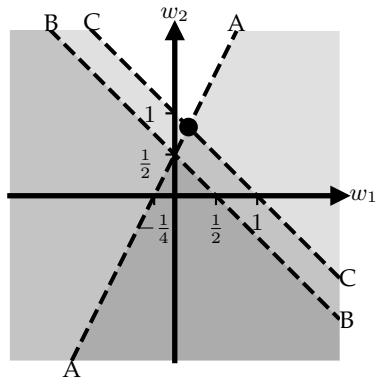
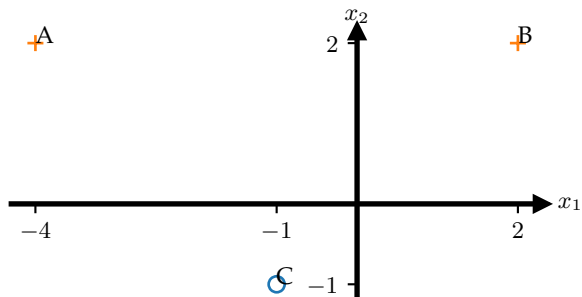
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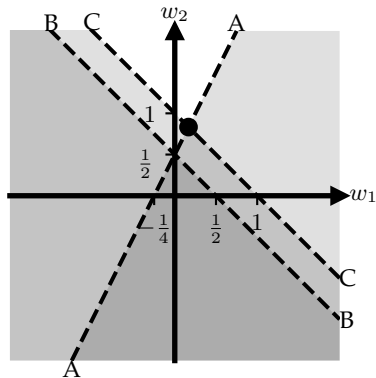
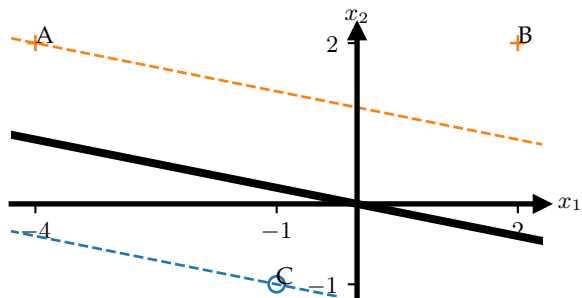
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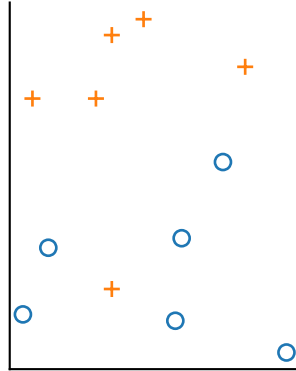
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Non-Separable Data

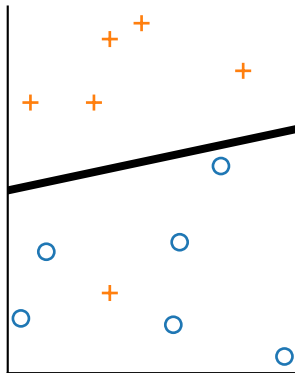
What if no margin is possible?



Non-Separable Data

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This seems reasonable

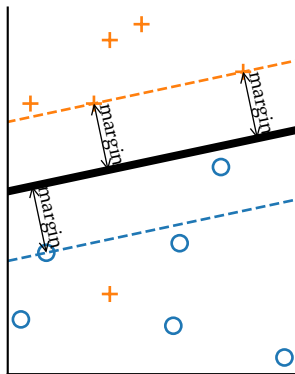


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Margins keep most points “away”



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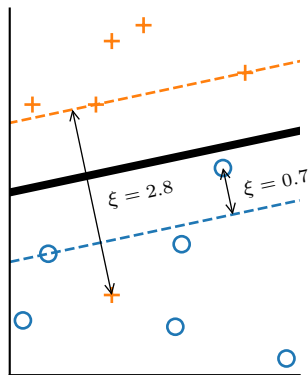
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Except a few...

ξ_i is the **slack variable** associated with point i



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opt vars

w/o slack variables $n + 1$

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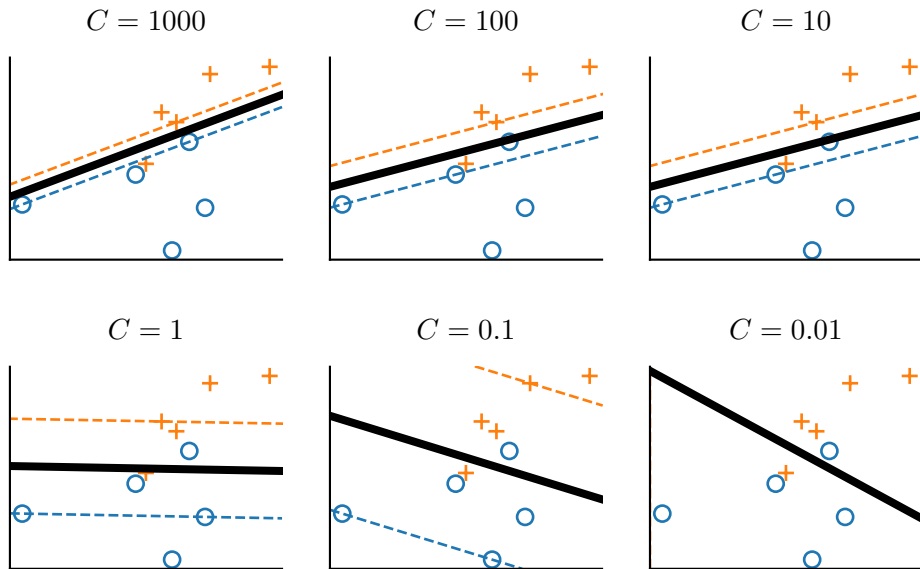
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	# opt vars	hyper-parameters
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w/o slack variables	$n + 1$	
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w/ slack variables	$n + m + 1$	C
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Effect of C



Is this so different?

$$\begin{array}{ll} \arg \min_{\vec{w}, b, \xi_1, \dots, \xi_m} & \frac{1}{2} \vec{w}^\top \vec{w} + C \sum_{i=1}^m \xi_i \\ \text{s.t.} & y_i (\vec{x}_i^\top \vec{w} + b) \geq 1 - \xi_i \quad \text{for all } i \in \{1, \dots, m\} \\ & \xi_i \geq 0 \quad \text{for all } i \in \{1, \dots, m\} \end{array}$$

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Is this so different?

$$\arg \min_{\vec{w}, b} \quad \frac{1}{2} \vec{w}^{\top} \vec{w} + C \sum_{i=1}^m \max(0, 1 - y_i (\vec{x}_i^{\top} \vec{w} + b))$$

SVM Loss Function

$$\arg \min_{\vec{w}, b} \frac{1}{2} \vec{w}^\top \vec{w} + C \sum_{i=1}^m \max(0, 1 - y_i (\vec{x}_i^\top \vec{w} + b))$$

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Is the same as

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Is the same as

$$\arg \min_{\vec{w}, b} \sum_{i=1}^m \max(0, 1 - y_i \tilde{f}(\vec{x}_i)) + \frac{1}{2C} \vec{w}^\top \vec{w}$$

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Is the same as

$$\arg \min_{\vec{w}, b} \sum_{i=1}^m l(y_i, \tilde{f}(\vec{x}_i)) + \lambda R(\vec{w})$$

where

$$l(y, \tilde{f}) = \max(0, 1 - y\tilde{f}) \quad R(\vec{w}) = \vec{w}^\top \vec{w} \quad \lambda = \frac{1}{2C}$$

SVM Loss Function

$$\arg \min_{\vec{w}, b} \frac{1}{2} \vec{w}^\top \vec{w} + C \sum_{i=1}^m \max(0, 1 - y_i (\vec{x}_i^\top \vec{w} + b))$$

Is the same as

$$\arg \min_{\vec{w}, b} \sum_{i=1}^m \max(0, 1 - y_i \tilde{f}(\vec{x}_i)) + \frac{1}{2C} \vec{w}^\top \vec{w}$$

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$$\arg \min_{\vec{w}, b} \sum_{i=1}^m l(y_i, \tilde{f}(\vec{x}_i)) + \lambda R(\vec{w})$$

where

$$l(y, \tilde{f}) = \overbrace{\max(0, 1 - y\tilde{f})}^{\text{the hinge loss}}$$

$$R(\vec{w}) = \vec{w}^\top \vec{w}$$

$$\lambda = \frac{1}{2C}$$

